MILITARY SPECIFICATION

KIT, CONVERSION
H4277/H4272 EXTRACTOR TOOL
M753 PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for extractor tool conversion kits used on M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GGG-W-636	Wrench (Box, Open End and Combination)

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1110

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MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MS51975	Screw, Shoulder-socket headed, Hexagon, Alloy Steel, Cadmium Plated, UNC-3A
MS171540	Pin, Spring, Corrosion Resistant Steel

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.7)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9327827	Kit, Conversion
9275089	Spring
9278459	Extractor Tool Subassembly, H4272, M753 Projectile
9278461	Bearing Plate
9278462	Forcing Cone Assembly
9299050	Motor Body, 8 Inch: M753
9312541	Rod, Rear
9312633	Tool, Holding, Bearing Plate
9312672	Extractor Tool Assembly, H4272, M753 Projectile
9321429	Screw, Shoulder
9321554	Jaw
9327745	Hammer, Non-sparking

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8	Tension Testing of Metallic Materials
ASTM E18	Tests for Rockwell Hardness of Metallic Materials
ASTM B117	Salt Spray (Fog) Testing
ASTM E138	Method for Wet Magnetic Particle Inspection
ASTM B567	Measurement of Coating Thickness by Beta Backscatter Method

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components. The contractor shall furnish objective evidence (see 6.4) that materials and components comply with the specifications listed on the applicable drawing (dwg).

3.2 Assembly. The conversion kit shall comply with all requirements specified on (dwg.) 9327827, with the requirements of all applicable subassembly dwgs., with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the dwgs. and herein.

3.3 Plating.

3.3.1 Quality. Plating finish shall be as specified on applicable dwgs. and shall be smooth, fine-grained and free of blisters, nodules, gouges, excessive build-up at corners or any other plating defects.

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3.3.2 Salt spray. Plating thickness shall be as specified on applicable dwgs., shall be adherent and shall meet the requirements of 3.3.1 after exposure to salt spray atmosphere for a minimum of 16 hours. Testing shall be performed in accordance with 4.5.1.

3.4 Performance.

3.4.1 Life cycle tensile strength. With the forcing cone end of the extraction tool subassembly restrained in a manner simulating usage (see dwg. 9312672), the extraction tool subassembly shall withstand a tensile force of 100,000 lb ± 10% exerted on the opposite rod end for 100 cycles at 70° ± 20°F without breakage, cracking, distortion or any other evidence of physical damage. Testing shall be performed in accordance with the applicable conditions of 4.5.2.

3.4.2 Tensile strength screening. Three cycles of tensile strength screening shall be performed on each extraction tool subassembly at 70° ± 20°F without breakage, cracking, distortion or any other evidence of physical damage. Testing shall be performed in accordance with the applicable conditions of 4.5.2.

3.5 Government furnished property. Unless otherwise indicated in the contract, the following items will be Government furnished:

Description	Drawing	Quantity
H4277 Mating Rod for tensile strength tests (see 4.5.2)	9312541	1

3.6 First article inspection. This specification contains technical requirements for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components and assemblies shall be free of cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity or any similar characteristics.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable dwg.

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3.7.3 Burrs. Parts and assemblies shall be free of burrs.

3.7.4 Foreign matter. Parts and assemblies shall be free of chips, dirt, grease, rust, corrosion or any embedded foreign material. Cleaning methods shall not be injurious to any of the parts and assemblies, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.6), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Item	Dwg. No.	Quantity
Kit, Conversion H4277/H4272	9327827	1

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4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwgs. and to the inspections and tests specified in the first article inspection table herein. All parts shall have been manufactured using the same materials, production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the same source as will be used in regular production. Inspections or tests required on parts and subassemblies before assembly shall be subjected to verification by a representative of the Government designated by the contracting officer.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-K-63185(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1	Kit, Conversion H4277/H4272				9327827
					NEXT HIGHER ASSEMBLY
					NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of Defects and Tests (see note 1)	1			4.4.2.2 thru 4.4.2.7

NOTES:

1. For first article tests, life cycle tensile strength per 3.4.1 and 4.5.2 shall be performed in lieu of the tensile strength screening specified in 4.4.2.6 (Major Category 106).

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4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot shall consist of items plated at the same time, with the same equipment, by the same plating process, with the same basic materials and shall not exceed one hundred items.

4.4.2 Examinations. See MIL-A-48078 and the quality conformance inspection tables herein.

a. Major defect sampling plans. Unless otherwise specified, inspection for Major defects shall be sampled in accordance with Table I.

b. Minor defect sampling plans. Unless otherwise specified, inspection for minor defects shall be sampled in accordance with Table II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-K-63185(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Commercial Components and MS Parts	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY: NA

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Objective evidence not available for the following items in accordance with the applicable dwg. or spec.		-	3.1	Visual (6.4)

Item	Dwg. or Spec.	Ref. on dwg.
Spring	9275089	9278459
Screw	MS51975-43	9278459
Pin, Spring	MS171540	9278459
Wrenches (2)	GGG-W-636	9327827
Hammer	9327745	9327827

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-K-63185(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Bearing Plate	1 OF 3	9278461
			NEXT HIGHER ASSEMBLY: 9278459

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Overall width (8.20 - .02)		Table I	3.2	Gage
102.	Location overall width				
103.	Flat width (1.87 - .01)		Table I	3.2	Gage
104.	Location flat width		Table I	3.2	Gage
105.	Flange width 4 places (1.50 - .06)		Table I	3.2	Gage
106.	Location flange width		Table I	3.2	Gage
107.	Pitch diameter internal thread		Table I	3.2	Gage
108.	Minor diameter internal thread		Table I	3.2	Gage
109.	Width of slot recesses 4 places (.89 + .02)		Table I	3.2	Gage
110.	Width of slots 4 places (.625 + .010)		Table I	3.2	Gage
111.	Length of slot recesses 4 places (1.30 + .04)		Table I	3.2	Gage
112.	Length of slot 4 places (1.00 + .02)		Table I	3.2	Gage
113.	Outside diameter (3.20 - .02)		Table I	3.2	Gage
114.	Outside diameter (2.25 - .03)		Table I	3.2	Gage
115.	Length of flats (.75 + .04)		Table I	3.2	Gage
116.	Overall width (3.30 - .03)		Table I	3.2	Gage
117.	Width (2.40 - .03)		Table I	3.2	Gage
118.	Width (1.90 - .03)		Table I	3.2	Gage
119.	Width (.25 - .02)		Table I	3.2	Gage
120.	Depth of slot (1.24 - .01)		Table I	3.2	Gage
121.	Depth of recess (1.25 + .02)		Table I	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-K-63185(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.2 (cont)	Bearing Plate				9278461
					NEXT HIGHER ASSEMBLY
					9278459
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major (cont)					
122.	Depth of recess (.10 + .02)		Table I	3.2	Gage
123.	Outside diameter (7.40 - .04)		Table I	3.2	Gage
124.	Inside diameter (6.64 + .04)		Table I	3.2	Gage
125.	Inside diameter (4.26 + .04)		Table I	3.2	Gage
126.	Location inside diameter		Table I	3.2	Gage
127.	Angle Recess (110 - 2)		Table I	3.2	Gage
128.	Angle Recess (120 + 2)		Table I	3.2	Gage
129.	Inside diameter recess 2.50 + .02)		Table I	3.2	Gage
130.	Depth recess (.50 + .01)		Table I	3.2	Gage
131.	Any camfer or radius missing		Table I	3.2	Visual
132.	Location 4 slot faces		Table I	3.2	Gage
133.	Location 4 slot recess faces		Table I	3.2	Gage
134.	Protective finish present with no bare spots		Table I	3.2	Visual
135.	Hardness		Table I	3.2	ASTM E18/Gage
136.	Yield strength	4.4. 3.1	-	3.2	ASTM E8/Gage
Minor					
201.	Surface finish 16, 125		Table II	3.2	Visual
202.	Marking incorrect, missing or incorrect location		Table II	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.2 (cont)	Bearing Plate		3 OF 3		9278461
					NEXT HIGHER ASSEMBLY 9278459
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor (cont)					
203.	Evidence of poor workmanship		Table II	3.7	Visual
204.	Material		-	3.1	Visual (6.4)

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-K-63185(AR)

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.3	Forcing Cone Assembly		9278462
			NEXT HIGHER ASSEMBLY 9278459

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Pitch diameter of 1.250 thread		100%	3.2	Gage
102.	Major diameter of 1.250 thread		100%	3.2	Gage
103.	Location of end of 1.250 thread		100%	3.2	Gage
104.	Length (.375 ± .010)		100%	3.2	Gage
105.	Outside diameter (1.14)		100%	3.2	Gage
106.	Width of grove in 2 places (.43 ± .04)		100%	3.2	Gage
107.	Location end of groove in 2 places (.69 ± .01)		100%	3.2	Gage
108.	Width of slots in 2 places (.30 ± .01)		100%	3.2	Gage
109.	Overall length of forward rod		Table I	3.2	Gage
110.	Length of flats (1.75 + .12)		Table I	3.2	Gage
111.	Location of end flat (5.18 ± .06)		Table I	3.2	Gage
112.	Width across flats (1.00 - .01)		Table I	3.2	Gage
113.	Outside diameter (1.25 - .01)		Table I	3.2	Gage
114.	Location of 1.25 diameter		Table I	3.2	Gage
115.	Pitch diameter 1.375 outside thread		Table I	3.2	Gage
116.	Major diameter 1.375 outside thread		Table I	3.2	Gage
117.	Length of 1.375 outside thread		Table I	3.2	Gage
118.	Inside diameter of pin hole		Table I	3.2	Gage
119.	Location of pin hole		Table I	3.2	Gage
120.	Outside diameter of groove (1.22 - .01)		Table I	3.2	Gage
121.	Location end of groove (.68 - .02)		Table I	3.2	Gage
122.	Location end of groove (1.17 + .02)		Table I	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-K-63185(AR)

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.3 (cont)	Forcing Cone Assembly		9278462
			NEXT HIGHER ASSEMBLY 9278459

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major (cont)					
123.	Inside diameter of cone cavity (3.62 + .01)		Table I	3.2	Gage
124.	Outside diameter of cone at reference location (4.604 - .01)		Table I	3.2	Gage
125.	Location of 4.604 diameter		Table I	3.2	Gage
126.	Correct taper on outside diameter of cone		Table I	3.2	Gage
127.	Overall cone width (2.70 - .02)		Table I	3.2	Gage
128.	Outside diameter of cone hub (2.30 - .02)		Table I	3.2	Gage
129.	Bevel width (.18 + .04)		Table I	3.2	Gage
130.	Bevel width (.06 + .03)		Table I	3.2	Gage
131.	Cone width (2.20 - .02)		Table I	3.2	Gage
132.	Minimum thickness cone at reference location		Table I	3.2	Gage
*133.	Pitch diameter cone thread		Table I	3.2	Gage
*134.	Minor diameter cone thread		Table I	3.2	Gage
135.	Location of end of rod in cone (.38 + .01)		Table I	3.2	Gage
136.	Any chamfer or radius missing		Table I	3.2	Visual
137.	Hardness		Table I	3.2	ASTM E18
*138.	Defective weld (magnetic particle inspection)		100%	3.2	ASTM E138

NOTES:

*Inspections preceded by an asterisk are not applicable to forced assemblies.

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PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.3 (cont)	Forcing Cone Assembly				9278462
					NEXT HIGHER ASSEMBLY 9278459
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major (cont)					
139.	Yield strength	4.4.3.1	-	3.2	ASTM E8
140.	Protective finish (plating quality)		100%	3.3.1	Visual
141.	Protective finish (salt spray)	4.4.3.2	-	3.3.2	4.5.1
Minor					
201.	Surface finishes 16, 125		Table II	3.2	Visual
202.	Marking improper or missing		Table II	3.2	Visual
203.	Evidence of poor workmanship		Table II	3.7	Visual
204.	Material		-	3.1	Visual (6.4)

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MIL-K-63185(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Tool, Holding, Bearing Plate		9312633
			NEXT HIGHER ASSEMBLY
			9327827

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Protective finish present with no bare spots		Table I	3.2	Visual
102.	Length of handle (46.98 - .04)		Table I	3.2	Gage
103.	Outside diameter socket (2.65 ± .010)		Table I	3.2	Gage
104.	Inside diameter socket (2.30 + .01)		Table I	3.2	Gage
105.	Width between flats (1.88 + .01)		Table I	3.2	Gage
106.	Width of flats (.90 min)		Table I	3.2	Gage
107.	Width of socket (1.13 - .02)		Table I	3.2	Gage
108.	Depth of recess		Table I	3.2	Gage
109.	Angle of bevel inside socket (46° - 2°)		Table I	3.2	Gage
110.	Inside diameter (2.510 + .002)		Table I	3.2	Gage
111.	Socket/handle welds per requirements (2 places)		Table I	3.2	Visual
112.	Perpendicularity ends (2)		Table I	3.2	Visual
Minor					
201.	Marking incorrect, missing or in incorrect location		Table II	3.2	Visual
202.	Evidence of poor workmanship		Table II	3.7	Visual
203.	Material		-	3.1	Visual (6.4)

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MIL-K-63185(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.5	Jaw				9321554
					NEXT HIGHER ASSEMBLY
					9278459
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Diameter outside taper at basic location		100%	3.2	Gage
102.	Outside taper correct		100%	3.2	Gage
103.	Overall width		Table I	3.2	Gage
104.	Location of groove from left face		Table I	3.2	Gage
105.	Width of groove		Table I	3.2	Gage
106.	Diameter inside taper at basic location		Table I	3.2	Gage
107.	Inside taper correct		Table I	3.2	Gage
108.	Outside diameter at bottom of groove		Table I	3.2	Gage
109.	Location of threaded hole		Table I	3.2	Gage
110.	Pitch diameter of thread		Table I	3.2	Gage
111.	Minor diameter of thread		Table I	3.2	Gage
112.	Bevel angle		Table I	3.2	Gage
113.	Bevel width		Table I	3.2	Gage
114.	Any chamfer, radius or countersink missing		Table I	3.2	Visual
115.	Depth of hole		Table I	3.2	Gage
116.	Depth of thread		Table I	3.2	Gage
117.	Protective finish (plating quality)		100%	3.3.1	Visual
118.	Protective finish (salt spray)	4.4.3.2	-	3.3.2	4.5.1
119.	Hardness	.	Table I	3.2	ASTM E18/Gage
120.	Yield strength and elongation	4.4.3.1	-	3.2	ASTM E8/Gage

NOTES:

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MIL-K-63185(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.5 (cont)	Jaw				9321554
					NEXT HIGHER ASSEMBLY
					9278459
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Surface finish 125, 16		Table II	3.2	Visual
202.	Marking incorrect, missing or in incorrect location		Table II	3.2	Visual
203.	Evidence of poor workmanship		Table II	3.7	Visual
204.	Material		Table II	3.1	Visual (6.4)

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-K-63185(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Extraction Tool Subasembly H4272, M753 Projectile	1 OF 1	9278459
			NEXT HIGHER ASSEMBLY 9327827

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Parts and surfaces clean prior to assembly		100%	3.2	Visual
102.	Threads coated with sealing compound prior to assembly		100%	3.2	Visual
103.	Assembly complete and correct		100%	3.2	Visual
104.	Spring pin projection (.19 ± .02)		100%	3.2	Gage
105.	Length of screw thread (see dwg. 9321429)		100%	3.2	Gage
106.	Tensile strength screening test at 70° + 20° F (3 cycles)		100%	3.4.2	4.5.2
Minor					
201.	Evidence of poor workmanship		Table II	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-K-63185(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.7	Kit, Conversion H4277/H4272		1 OF 1		9327827
					NEXT HIGHER ASSEMBLY: NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Improper or missing part or subassembly		100%	3.2	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Packaging				See 5.0
					NEXT HIGHER ASSEMBLY
					NA
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Packaging, packing or marking not in accordance with applicable requirements		-	5.0	Visual
102.	Any part or subassembly missing (see dwg. 9327827)		100%	5.0	Visual
103.	Items not clean and free of dirt and contamination prior to packaging		Table I	5.0	Visual
104.	Items not individually packed (when required)		Table I	5.0	Visual
105.	Items not cushioned and firmly packed		Table I	5.0	Visual/Manual
Minor					
201.	Marking missing, incorrect or illegible		Table II	5.0	Visual

NOTES:

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables and herein.

4.4.3.1 Yield strength. Specimens of 10% of each heat treat lot of materials used to fabricate the parts shall be tested for yield strength. Unless otherwise specified on dwgs., the specimens may be of any convenient size or configuration which have been heat treated with the same lot regardless of heat treat procedure (single, batch and/or continuous). Testing shall be in accordance with ASTM E8 or equivalent. If any specimen fails to comply with the yield strength requirements of the applicable dwg., the heat treat lot of parts shall be rejected.

4.4.3.2 Plating (salt spray). Specimens from each plating inspection lot (see 4.4.1) plated concurrently with the actual items shall be tested for the applicable requirements in accordance with 4.5.1. The specimens shall be prepared from the same base material, in essentially the same configuration, with the same surface finish and shall be cleaned and processed concurrently with the items to be plated. The test specimens shall be at least three in number and located in positions in the plating process which will register thickness and quality representative of the plating process. If any sample specimen fails to meet any of the applicable plating requirements, the plating inspection lot shall be rejected.

Table I. Sampling plan for major defects.

LQ = .05 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table II. Sampling plan for minor defects.

LQ = .10 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see MIL-A-48078 and 6.3 herein).

4.5 Test methods and procedures.

4.5.1 Salt spray (see 3.3.2). Salt spray tests shall be conducted in accordance with ASTM B117 or an approved equivalent. The following details and exceptions shall apply:

a. Salt spray exposure time shall be 16 hours.

b. Following the salt spray exposure, plating shall be visually examined for the requirements of 3.3.1, plating thickness shall be measured in accordance with ASTM B567 or other approved method, and plating adhesion examined. Plating adhesion shall be examined by shearing the plating down to base metal with a suitable sharp cutting tool and examining the plating/base metal adhering surfaces under 4x magnification for any evidence of plating/base metal separation.

4.5.2 Tensile strength (see 3.4). The forcing cone end of the extraction tool subassembly shall be fully expanded in the boattail section of a restrained rocket motor body (see dwg. 9299050) or approved equivalent in a manner simulating actual use conditions (see dwg. 9312672). This is not to be construed literally as meaning an actual rocket motor body or cannon breech or breechblock assembly are required to perform

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these tests or that the tests need be performed horizontally. Unless otherwise specified in the contract, Government approved, contractor designed and furnished fixtures simulating these items shall be used and at the contractors option testing may be performed vertically. An H4277 mating rod (dwg 9312541) furnished by the acquiring activity shall be threaded on the forward rod end of the H4272 subassembly and secured by a retaining bolt in a manner simulating actual use.An approved forcing mechanism shall be secured to the mating rod. The applicable tensile force indicated below shall be exerted on the mating rod, retained for 10 seconds minimum, and released. The tensile force shall be repeated for the number of cycles indicated. After being subjected to the required number of cycles, the H4272 extraction tool subassembly shall be visually examined for evidence of physical damage including inspection of the H4272 rod thread with approved thread gages.

First Article Life Cycle Test:

100,000 lb at 70° ± 20°F (100 cycles)

Quality Conformance Tensile Strength Screening:

75,000 lb at 70° ± 20°F (3 cycles)

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified in the contract, assemblies shall be packaged, packed and marked in wooden boxes in accordance with the requirements of dwg. 9322197.

6. NOTES

6.1 Intended use. This specification contains requirements for the H4272 Conversion Kit which includes the M753 Projectile Extraction Tool Subassembly and all necessary ancillary hardware. This conversion kit will be shipped to the field and depots for inclusion in the H4277/H4272 Extractor Atomic Projectile already fielded for M422 use.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Government furnished propery (see 6.5).
c. First article inspection, when required (see 3.6).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence.

6.4.1 General. Records of contractor quality controls and inspections which can be verified.

6.4.2 Source control dwg., spec. control dwgs., MIL-SPEC items and MS parts. A certificate of conformance from the supplier certifying that the item complies with all dwg. or specification requirements.

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6.4.3 Materials. A statement (certification) supported by test data that all materials produced or purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any contractual documents.

6.5 Government furnished property. The contracting officer should arrange to furnish the property listed in 3.5.

6.6 Supersession data. This specification includes the requirements of AR-PD-2 dated 11 September 1978 with Amendment 2 dated 4 December 1980.

6.7 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading US Army Armament Research and Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:

Army-AR

Preparing activity:

Army-AR

(Project 1110-A166)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-K-63185 KIT, CONVERSION H4277/H4272 EXTRACTOR TOOL M753 PROJECTILE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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